ANSWER KEY & MARKING SCHEME SET A

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 84 cm²	$l \times b$
2	**(d)** 30 cm²	$\frac{1}{2} \times b \times h$
3	**(d)** 54 cm²	$\frac{1}{2} \times b \times h$
4	**(a)** 120 cm²	base × height
5	**(b)** 90 cm²	$\frac{1}{2}(a+b)h$
6	**(b)** 96 cm²	$\frac{1}{2}d_1d_2$
7	**(a)** 12 cm	Area ÷ length
8	**(d)** 10000 cm²	100 × 100
9	**(b)** equal area	Same base and height
10	**(c)** 81 cm²	Side 9 cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{2}bh$ (1); bh (1).	1+1
12	198 m² (2).	2
13	Different shapes with equal perimeter can have different areas (1); e.g. 6×2 and 4×4 rectangles (1).	1+1
14	60 cm² (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area 126 m² (1.5); perimeter 46 m, cost ₹ 1380 (1.5).	1½+1½
16	Area 260 m² (1.5); perimeter 66 m, cost ₹ 1980 (1.5).	1½+1½
17	Area 200 m² (1.5); perimeter 66 m, cost ₹ 1980 (1.5).	1½+1½
	OR: Area 480 m² (1.5); perimeter 94 m, cost ₹ 2820 (1.5).	1½+1½
18	Area 324 m² (1.5); perimeter 72 m, cost ₹ 2160 (1.5).	1½+1½
	OR: Area 162 m² (1.5); perimeter 66 m, cost ₹ 1980 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A diagonal halves a rectangle; or two congruent triangles make a parallelogram (3). (ii) $\frac{1}{2} \times 14 \times 12 = 84$ cm² (2).	3+2
	OR: (i) 3000 m² (2). (ii) ₹ 6000 (1). (iii) 0.3 ha (2).	2+1+2
20	(i) $\frac{1}{2}ah + \frac{1}{2}bh$ (3). (ii) 300 cm² (2).	3+2
	OR: (i) Equal bases and common height (3). (ii) 40 cm² (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 120 cm²	$l \times b$
2	**(a)** 63 cm²	$\frac{1}{2} \times b \times h$
3	**(a)** 60 cm²	base × height
4	**(a)** 72 cm²	$\frac{1}{2}(a+b)h$
5	**(d)** 135 cm²	$\frac{1}{2}(a+b)h$
6	**(c)** 126 cm²	$\frac{1}{2}d_1d_2$
7	**(b)** 8 cm	Area ÷ length
8	**(a)** equal area	Congruent
9	**(d)** doubles	Area ∝ base
10	**(d)** 6 cm	72 ÷ 12

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	77 cm² (2).	2
12	96 cm² (2).	2
13	12 cm (2).	2
14	$200000/500 = 400$ tiles (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Trapezium $= \frac{1}{2}(12+20) \times 8 = 128$ cm² (2); rectangle 160 cm² (1).	2+1
16	Trapezium $= \frac{1}{2}(9+15) \times 10 = 120$ cm² (2); rectangle 150 cm² (1).	2+1
17	Trapezium $= \frac{1}{2}(14+22) \times 7 = 126$ cm² (2); rectangle 154 cm² (1).	2+1
	OR: Trapezium $= \frac{1}{2}(18+30) \times 12 = 288$ cm² (2); rectangle 360 cm² (1).	2+1
18	Trapezium $= \frac{1}{2}(10+26) \times 9 = 162$ cm² (2); rectangle 234 cm² (1).	2+1
	OR: Trapezium $= \frac{1}{2}(16+24) \times 15 = 300$ cm² (2); rectangle 360 cm² (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 216 cm² (2). (ii) 15 cm (1). (iii) $54000/216 = 250$ tiles (2).	2+1+2
	OR: (i) $1.2 \times 0.8 = 0.96$, a 4% decrease (3). (ii) Changes multiply (2).	3+2
20	(i) $\frac{1}{2} \times 20 \times (7+9) = 160$ cm² (3). (ii) Dividing into two triangles on the diagonal (2).	3+2
	OR: (i) $10 \times 6 = 60$ m² (2). (ii) $96 - 60 = 36$ m² (1). (iii) ₹ 1980 (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.